

Anchor-Symmetry Error Certificate

Feedback-driven rerun | 21 September 2026

Calibration-only theory candidates, rebuilt spectra, and trial-level uncertainty. This supplement reports a new exploratory audit; it does not overwrite the historical figures.

1. Canonical result: 192 mW law ceiling

Three test trials: 250, 300 and 350 mW. Twelve low-power training trials; four above-break law-calibration trials. The table retains the original LOTO anchor-error heuristic for a comparable allowance ablation.

Allowance	Covered trials	CP lower	S > 1	Mean S/e	Min margin
Fitted rate	2/3	0.135	1.000	0.468	-0.298
Historical 2x	3/3	0.368	1.000	0.153	2.114
Theory slope	3/3	0.368	1.000	0.372	0.438
Theory + boundary	3/3	0.368	1.000	0.298	1.345

2. All 22 splits and all 12 training anchors

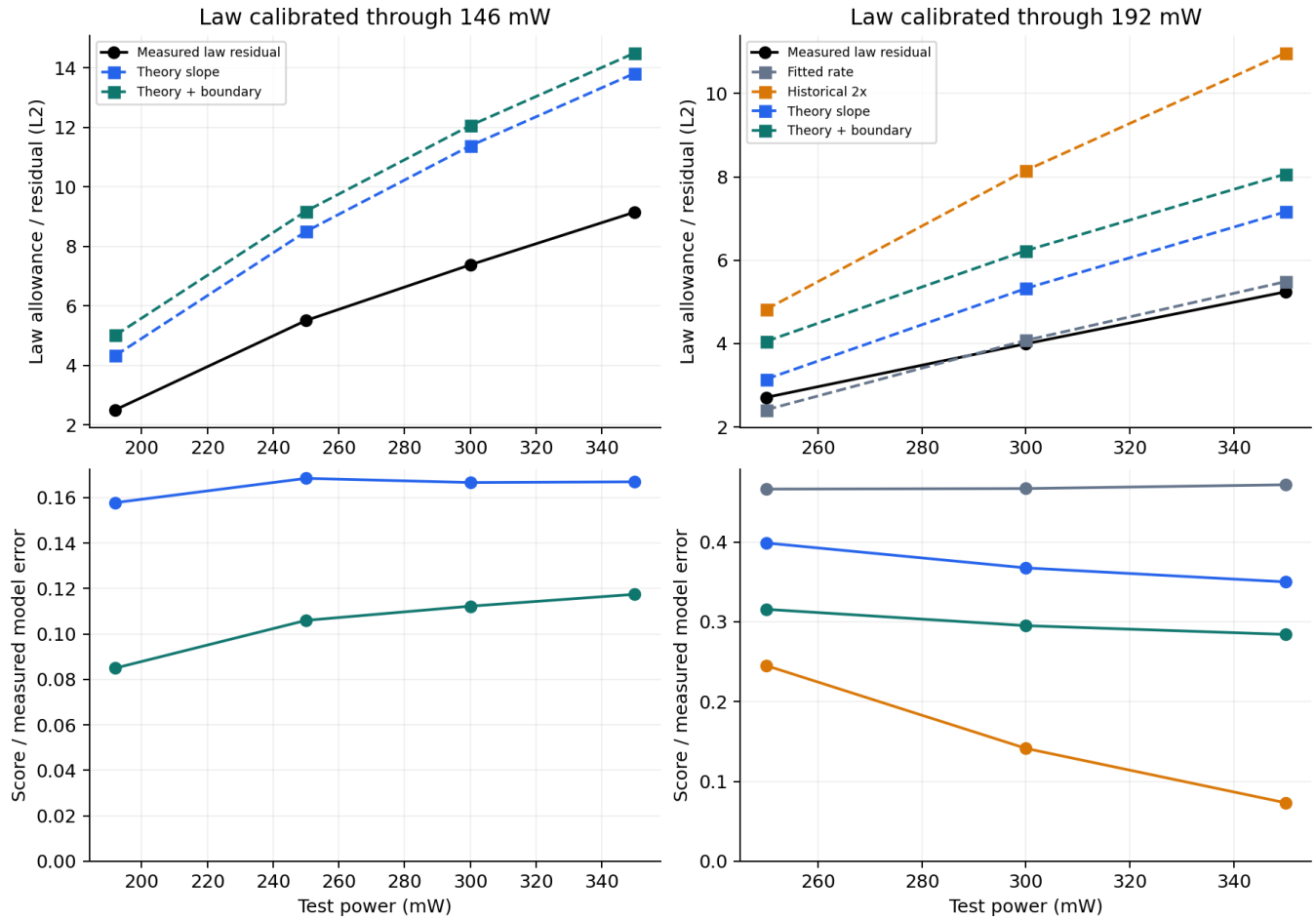
348 configuration rows reuse only three distinct test trials. Pooled fractions are descriptive. The count/CP columns below instead require each trial to pass every swept configuration; they do not form a confidence interval for the pooled rate.

Allowance	Pooled coverage	All-config trials	CP lower	S > 1	Mean S/e
Fitted rate	0.264	0/3	0.000	0.994	0.362
Historical 2x	0.925	1/3	0.017	0.753	0.196
Theory slope	0.296	0/3	0.000	0.991	0.402
Theory + boundary	1.000	3/3	0.368	0.848	0.240

The review's 0.21 baseline is reproduced at the fixed 54 mW anchor: 6/29 configuration rows (0.207; 3 distinct trials, all-config CP lower 0.000). The 0.264 baseline above additionally sweeps anchors.

No row proves 90% population coverage: even 3/3 only gives a nominal lower bound of 0.368. The historical 2x rule was informed by earlier inspection of these test data; it remains a sensitivity comparator.

3. What changes at each test power



Same trained MLP(8), seed 0, and legacy LOTO b throughout. The 146 mW configuration has four distinct test trials; the Setup C fitted rule cannot be estimated there because it requires a three-trial sub-law plus a holdout.

Ceiling	Theory k	Boundary norm	Covered trials	CP lower	Mean S/e
146 mW	15.797	0.682	4/4	0.473	0.105
192 mW	11.926	0.906	3/3	0.368	0.298

The boundary value comes from the already-labelled calibration ceiling and anchor. No test residual enters this value or the theory slope.

4. Separate law adequacy from anchor calibration

The legacy b uses residuals from different leave-one-trial-out models. Its order statistic does not establish split-conformal coverage for the final model at a selected anchor. The labelled-anchor variant instead uses the measured final-model error at that already-labelled anchor. This is exact for the empirical trial-mean target only.

Model / seed	b: LOTO / anchor	Mean S/e: LOTO / anchor	Valid trials: LOTO / anchor
mlp8 / 0	7.003 / 1.487	0.298 / 0.747	3/3 / 3/3
mlp8 / 1	6.970 / 1.475	0.274 / 0.739	3/3 / 3/3
mlp8 / 2	7.144 / 1.588	0.260 / 0.735	3/3 / 3/3
mlp16x16 / 0	5.632 / 1.394	0.000 / 0.000	3/3 / 3/3
mlp16x16 / 1	4.976 / 1.271	0.000 / 0.008	3/3 / 3/3
mlp16x16 / 2	4.752 / 1.458	0.150 / 0.597	3/3 / 3/3
linear / 0	6.366 / 0.721	0.238 / 0.637	3/3 / 3/3

Each validity fraction of 3/3 corresponds to a nominal 95% CP lower bound of 0.368, not seven independent replications of three trials. Law coverage does not depend on model or seed; model sensitivity changes informativeness.

Conditional theory, not a universal guarantee

The capillary spectrum scales as $\text{energy flux}^{(1/2)}$ and $\text{frequency}^{(-17/6)}$. The relevant power slope is 1/2 only under energy flux proportional to drive power. The frequency exponent must not be compared to fitted power exponents.

For $u = \ln(\text{power})$, $r(u) = y_{\text{anchor}} - y(u) - c(u) - c_0$. If $\text{norm}(y'(u) - q_{\text{high}}) \leq k$ over the extrapolation interval, then $\text{norm}(r(u)) \leq \text{norm}(r(u_0)) + k*(u-u_0)$. We test $k = \text{norm}(q_{\text{high}} - 0.5*\text{ones})$.

The derivative assumption is not established by weak-turbulence theory for this apparatus. The experiment audits that candidate; it cannot upgrade it to an unconditional physics or coverage theorem.

5. Paper corrections and related work

Narrow the gap: physics-residual conformal methods can use no target labels. Our distinction is a per-input error lower bound with separate anchor and approximate-law allowances.

Separate training shift from calibration/test exchangeability. OOD relative to training does not automatically invalidate conformal calibration.

Report 348 configurations and three distinct test trials together. A pooled rate over overlapping configurations has no ordinary binomial confidence interval.

Describe the factor of two as retrospective. A round number chosen after viewing test residuals is still test-informed.

Correct reference [7] to Weak turbulence of capillary waves. Correct the first interior 90% finite-sample quantile to $n=19$, not 20. Neither rank arithmetic nor LOTO residuals alone establish exchangeability.

Gopakumar et al. Physics-informed UQ: label-free residual-space calibration. [Source](#)

Wang et al. Symmetry mismatch already yields regression error lower bounds. [Source](#)

Gopakumar et al. Surrogate conformal UQ retains calibration/test exchangeability. [Source](#)

Berman et al. Calibration-error bounds for equivariant functions. [Source](#)

Xu et al. Branched normalizing flows and conditional conformal transport. [Source](#)

Lust and Condurache. GIT transformation-based classification error detection. [Source](#)

Zakharov and Filonenko. Weak turbulence of capillary waves (1967). [Source](#)

Kochurin and Russkikh. Explicit flux and frequency scaling, equations 1-2. [Source](#)

Replacement paragraphs, assumptions and citation distinctions are in PAPER_REVISIONS.md. They do not adopt the reviewer's suggested acceptance-score estimates.

6. Reproducibility and artifact

All rebuilt dataset arrays exactly match the archived arrays: True. Maximum absolute log-spectrum difference: 0.

The source is the supplied 50 MB capillary-wave HDF5, derived from the Orosco, Connacher and Friend UC San Diego deposit (DOI 10.6075/J0WW7HVJ). The repackaging is lossy; this run does not recreate the original 320.7 GB reduction.

Target: average of window-level natural log band-averaged temporal PSDs. Twelve log-spaced bands, 500 Hz-20 kHz; 8192-sample windows, 4096 stride; Welch segments 2048, half overlap, Hann window, linear detrending. All model/law/test partitions are grouped by trial.

Artifact	Contents
RERUN_PROTOCOL.md	Fixed rules and assumptions written before candidate audit
theory_audit.py	Calibration fits, model reruns and row-level audit
rerun/calibration_fits.json	266 fitted configurations, trial lists and frozen allowances
rerun/audit_rows.csv	3176 rows including abstentions, margins and both premises
rerun/results.json	Summary, versions, input/code hashes and rebuild comparison
test_theory_audit.py	Five tests: leakage, boundary, trial counts, abstention and inequality
README_RERUN.md	Commands and interpretation
PAPER_REVISIONS.md	Supported replacement prose and source links

Python 3.8.10; numpy 1.24.4, scipy 1.10.1, scikit-learn 1.3.2, h5py 3.11.0, matplotlib 3.7.5, reportlab 3.6.13

Scope: two canonical ceilings, a 22-split x 12-anchor sensitivity sweep, seven model/seed fits, four fixed allowance rules, two anchor-error choices, and five new verification tests. Historical ensemble benchmarks and earlier figure atlases are not represented as newly rerun results.

Limits: only three or four distinct test trials, one apparatus, already-viewed test data, uncertain drive-to-flux coupling, and empirical trial means rather than a noise-free physical target. Additional independent trials are required to assess the 0.90 population-coverage claim.